1. PAC Learning Framework

1. Setup

Components:

- Input space $\mathcal{X}$
- Label space $\mathcal{Y} = \{0, 1\}$
- True concept $c : \mathcal{X} \to \mathcal{Y}$
- Concept class $\mathcal{C}$
- Hypothesis set $\mathcal{H}$
- Distribution $\mathcal{D}$ over $\mathcal{X}$
- Sample $S = \{(x_i, y_i)\}_{i=1}^m$

Error measures:

- Generalization error: $R(h) = \mathbb{P}_{x\sim\mathcal{D}}[h(x) \neq c(x)]$
- Empirical error: $R_{\text{emp}}(h) = \frac{1}{m}\sum_{i=1}^m \mathbb{1}_{h(x_i)\neq y_i}$

2. PAC Learnability

$\mathcal{H}$ is **PAC-learnable** if exists algorithm A and $m_{\mathcal{H}}(\varepsilon, \delta)$ such that:

For $m \geq m_{\mathcal{H}}(\varepsilon, \delta)$,

$$\mathbb{P}_S[R(A(S)) \leq \varepsilon] \geq 1 - \delta$$

Dimensions:

- **Probably** $(1 - \delta)$: high confidence
- **Approximately** (ε): small error
- **Correct**: low generalization error

2. Finite Hypothesis Sets

1. Realizable Case

Assumption: $\exists h^* \in \mathcal{H}$ with $R(h^*) = 0$.

Algorithm: Return any h_S with $R_{\text{emp}}(h_S) = 0$.

Sample complexity:

$$m \geq \frac{1}{\varepsilon}\left(\log|\mathcal{H}| + \log\frac{1}{\delta}\right)$$

Key insight: Logarithmic in $|\mathcal{H}|$.

2. Agnostic Case

No perfect hypothesis assumed.

Algorithm: Empirical Risk Minimization (ERM):

$$h_S = \arg\min_{h\in\mathcal{H}} R_{\text{emp}}(h)$$

Sample complexity:

$$m \geq \frac{2}{\varepsilon^2}\left(\log|\mathcal{H}| + \log\frac{2}{\delta}\right)$$

Key difference: $O(1/\varepsilon^2)$ vs $O(1/\varepsilon)$.

3. Growth Functions

1. Definitions

Dichotomy: Restriction of $\mathcal{H}$ to finite set $C \subset \mathcal{X}$:

$$\mathcal{H}_C = \{(h(x_1), \ldots, h(x_n)) : h \in \mathcal{H}\}$$

Growth function:

$$\Pi_{\mathcal{H}}(m) = \max_{C\subset\mathcal{X},\ |C|=m} |\mathcal{H}_C|$$

Upper bound: $\Pi_{\mathcal{H}}(m) \leq 2^m$

Shattering: $\mathcal{H}$ shatters C if $|\mathcal{H}_C| = 2^{|C|}$ (all labelings possible).

2. Key Properties

Monotonicity: $\Pi_{\mathcal{H}}(m)$ is non-decreasing in m.

Binary dichotomies: $\Pi_{\mathcal{H}}(m)$ counts distinct behaviors on size-m sets.

Uniform convergence: Growth function controls uniform deviation between R_{emp} and R.

4. VC Dimension

1. Definition

VC dimension $d = \text{VCdim}(\mathcal{H})$ is the largest m such that $\Pi_{\mathcal{H}}(m) = 2^m$.

Equivalently: largest set size that $\mathcal{H}$ can shatter.

If $\mathcal{H}$ can shatter arbitrarily large sets: $\text{VCdim}(\mathcal{H}) = \infty$.

2. Examples

Intervals on $\mathbb{R}$:

$$\mathcal{H} = \{h_{a,b} : h_{a,b}(x) = \mathbb{1}_{a\leq x\leq b}\}$$

$\text{VCdim}(\mathcal{H}) = 2$

Linear classifiers in $\mathbb{R}^d$:

$$\mathcal{H} = \{h_w : h_w(x) = \text{sign}(\langle w, x\rangle)\}$$

$\text{VCdim}(\mathcal{H}) = d$

Axis-aligned rectangles in $\mathbb{R}^2$: $\text{VCdim}(\mathcal{H}) = 4$

3. Computing VC Dimension

Strategy:

1. **Lower bound:** Find a set of size d that is shattered
2. **Upper bound:** Prove no set of size $d + 1$ can be shattered

Tip: For geometric classes, try “general position” configurations.

5. Sauer's Lemma

1. Statement

If $\text{VCdim}(\mathcal{H}) = d < \infty$, then:

$$\Pi_{\mathcal{H}}(m) \leq \sum_{i=0}^{d}\binom{m}{i} \leq \left(\frac{em}{d}\right)^d$$

Key transition: Exponential $(2^m) \to$ Polynomial $(O(m^d))$

When $m > d$: Growth becomes polynomial, not exponential.

2. Implications

Finite VC dimension $\Rightarrow$ polynomial growth

This is the key to PAC learnability for infinite hypothesis classes!

Critical threshold: $m = d$ (VC dimension)

6. VC Bound

1. Main Theorem

If $\text{VCdim}(\mathcal{H}) = d < \infty$, then $\mathcal{H}$ is PAC-learnable with:

$$m \geq \frac{c}{\varepsilon^2}\left(d + \log\frac{1}{\delta}\right)$$

for some constant c.

Sample complexity: Linear in d, not $\log|\mathcal{H}|$.

Algorithm: ERM over $\mathcal{H}$.

2. Comparison

| Setting | Bound | |———|——-| | Finite $\mathcal{H}$ (realizable) | $O\left(\frac{\log|\mathcal{H}|}{\varepsilon}\right)$ | | Finite $\mathcal{H}$ (agnostic) | $O\left(\frac{\log|\mathcal{H}|}{\varepsilon^2}\right)$ | | Infinite $\mathcal{H}$, VC=d | $O\left(\frac{d}{\varepsilon^2}\right)$ |

Backbone: From counting hypotheses to counting behaviors.

7. Key Pitfalls

VC $\neq$ num parameters: VC dimension is about shattering capacity, not model size. A 10-parameter model can have infinite VC dimension.

Shattering $\neq$ fitting: Just because $\mathcal{H}$ fits some m-point set doesn't mean it shatters it. Must fit **all** 2^m labelings.

Realizable vs agnostic: Realizable assumes perfect h^* exists; agnostic doesn't. Sample complexity differs by ε vs ε^2.

Union bound: Used to control probability over **all** bad hypotheses. Key technique in PAC bounds.

$m > d$ doesn't guarantee shattering: Sauer says $\Pi_{\mathcal{H}}(m)$ becomes polynomial, not that shattering stops at exactly d.